Syllabus

BTS 510 Syllabus – Fall 2026

Instructor information

- Stefany Coxe, Ph.D.
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  - Office hours: By appointment (send me an email to schedule)
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Course information

Learning goals

This course covers *linear models* (i.e., regression) with an emphasis on **experimental studies**. Topics include using and manipulating datasets, plotting data, estimation and uncertainty, linear regression, logistic regression, Poisson regression, and survival analysis.

Learning objectives

- **Select** the appropriate linear model for the research question
- **Analyze** data with the appropriate linear model
- **Describe** your findings for scientific and non-scientific audiences
- **Create** a written report of your findings
- **Make conclusions** about your research question(s) based on those results

Course structure

Time and location

We will meet in-person on **Thursday** from *10am to 11:30am* in **BTS classroom on the 7th floor of PDC**.

Course format

This course takes place in a **flipped, hybrid format**. We will meet in person for 1 hour 30 minutes each week. This time will be spent on **hands-on statistical programming in R**. You will be responsible for completing course assignments such as videos and readings **prior to class** in order to be prepared to participate in the in-person meeting. You will also have assignments to complete after class.

Each week will follow a similar structure:

- Monday: Lecture videos posted
- Wednesday: Watch lecture and respond to survey by 8pm Wednesday
- Thursday: In-person meeting to review material and work on applications
- Sunday: Homework assignment due by end of day (midnight)

Software

We will be using **R** for this course. It's ok if you don't know how to use R (but also great if you do!). We'll start from scratch in the first few weeks. I will provide information about the specific procedures you will need to know for this course.

- R is free and open source and works on any platform (Windows, Mac, Unix)
  - Download R here
  - I also recommend using Rstudio, which can be downloaded here
  - If you use a Chromebook or tablet, you can use Rstudio via the cloud
    - I have not used this much, so I don't know all its shortcomings
- We'll also be using Quarto
  - Publishing system based on markdown
  - Intersperse plain text and code
  - Output to convenient formats, like HTML, PDF, Word
  - Install Quarto

Assessments

Your work in this course will be assessed using a variety of methods.

Lecture videos (10%)

Watch the lecture video. Respond to the survey afterward with questions and comments.

Homework (60%)

There will be **five (5)** homework assignments. The assignments generally involve running analyses in R, interpreting output, and presenting the results.

Final project (30%)

You will select a dataset, propose some research questions that can be answered using linear models, write up the results, and present your findings to the class. More details to follow throughout the semester.

- Dataset selection: 5% of project grade
- Proposal: 15% of project grade
- Presentation: 40% of project grade
- Final paper: 40% of project grade

Tentative schedule

Week	Date	Topic	L	H
1	Sept 03	Introduction, R, markdown	1	
2	Sept 10	Data manipulation	2	1
3	Sept 17	Samples, sampling distribution	3	
4	Sept 24	ggplot	4	Data
5	Oct 01	ggplot	5	2
6	Oct 08	Linear regression	6	
7	Oct 15	Linear regression	7	3
8	Oct 22	Linear regression	8	
9	Oct 29	Linear regression	9	Proposal
10	Nov 05	Linear regression	10	4
11	Nov 12	Logistic regression	11	
12	Nov 19	Poisson regression	12	5
13	Nov 26	NO CLASS		
14	Dec 03	Survival analysis	13	

Week	Date	Topic	L	H
15	Dec 10	Repeated measures	14	
16	Dec 17	Presentations		Final

- **L**: Watch lecture videos and respond to survey by **Wednesday at 8pm**.
- **H**: Homework assignments are due by the **end of Sunday (midnight)**.
- **Final project**
  - **Data**: Description of data due by September 27
  - **Proposal**: Proposal of final project due by November 1
  - **Presentation**: Presentation in class on December 17
  - **Paper**: Final paper due by December 20
- **All due dates subject to change** due to weather, emergency, scheduling changes, etc.

Grades

Grade	Percentage
A+	>=97
A	93 - 96.99
A-	90 - 92.99
B+	87 - 89.99
B	83 - 86.99
B-	80 - 82.99
C+	77 - 79.99
C	70 - 76.99
D+	65 - 69.99
D	60 - 64.99
F	<= 59.99

Cedars-Sinai policies

Attendance

Attendance is not explicitly part of your grade in this course, but activities completed during the in-person portion of the course will be **very** helpful. If you need to miss class (such as for illness, religious event, professional activity, university-sanctioned event, or **any other reason**), please contact me to make any necessary arrangements.

Academic dishonesty and misconduct

Please refer to your policy handbook for a description of what constitutes academic dishonesty.

- **While you may work with other students on your homework assignments, I expect all students to complete and turn in their own work.**